

Department of Physics, FNSPE, CTU in Prague
02YMECH - Mechanics
Midterm 2

19.12.2025 / 14:00 - 15:30

Q1: Consider a system of three point particles, each having an identical mass m , with their positions defined by time-dependent vectors,

$$\vec{r}_1 = (at^2, vt, 0), \vec{r}_2 = (vt, at^2, b), \vec{r}_3 = (0, b, at^2),$$

where a , b , and v are positive constants.

- a. Find the total force required to produce the motion. — *0.2 points*
- b. Find the total angular momentum and torque concerning the origin of the coordinate system. — *0.3 points*
- c. Based on the results from parts (a) and (b), determine if total linear momentum and total angular momentum are conserved. — *0.2 points*
- d. Find the total kinetic energy in the LAB frame and in the center of mass frame. — *0.3 points*

Q2: The equation of motion of a damped oscillator is given as $x(t) = e^{-\beta t} x_0 \cos(\omega_1 t)$. How long does it take for the damped oscillator to halve the amplitude of the oscillation? (Givens: $\ln 2 \approx 0.7$, $m = 1\text{kg}$, $b = 1.4\text{ kg/s}$) — *1 point*

Q3: For a given central force $f(r)$ the orbit of a particle of mass μ can be calculated by integrating the following differential equation,

$$\frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) + \frac{1}{r} = -\frac{\mu}{l^2} r^2 f(r)$$

where l is the angular momentum of the particle. The particle follows an orbit given by $1 + \epsilon \cos \theta = \alpha/r$, where $\alpha \equiv \frac{l^2}{\mu k}$ and $\epsilon \equiv \sqrt{1 + \frac{2El^2}{\mu k^2}}$ are constants.

- a. Find the force law $f(r)$ for the central force in terms of k and r . — *0.6 points*
- b. Find the potential $U(r)$ associated with $f(r)$. — *0.4 points*

Q4: The earth rotates about its axis with angular velocity $\vec{\omega}$. The unit vectors $\hat{e}_1$, $\hat{e}_2$, and $\hat{e}_3$ are local unit vectors on the Earth's surface, where $\hat{e}_3$ is perpendicular to the surface, while $\hat{e}_1$ and $\hat{e}_2$ lie on the surface. The component of $\vec{\omega}$ along the $\hat{e}_3$ direction is ω_z . A very fast particle of mass m is projected on a horizontal surface with an initial velocity $\vec{v}_0 = v_0 \hat{e}_2$. Assume the Earth is a perfect sphere with radius R and neglect air resistance.

- a. Determine the Coriolis force by considering the general velocity of the particle $\vec{v} = v_1 \hat{e}_1 + v_2 \hat{e}_2$. Describe qualitatively the deflection due to the Coriolis force. — *0.6 points*
- b. Write the coupled equations of motion for $\hat{e}_1$ and $\hat{e}_2$ directions, and solve them for v_1 and v_2 . Does this result satisfy your initial expectations for the trajectory? — *0.4 points*

(Given: Coriolis force $\vec{F}_c = -2m\vec{\Omega} \times \vec{v}$)